

In []:

Numpy crash course

In [1]: `import numpy as np`

In [2]: `np.__version__`

Out[2]: `'2.3.5'`

In []:

In [3]: `My_list = [0,1,2,3,4,5]`
`My_list`

Out[3]: `[0, 1, 2, 3, 4, 5]`

In [5]: `arr = np.array(My_list)`
`arr`

Out[5]: `array([0, 1, 2, 3, 4, 5])`

In [6]: `print(type(My_list))`
`print(type(arr))`

`<class 'list'>`
`<class 'numpy.ndarray'>`

Arange function

In [7]: `np.arange(10)` *# here apply (n-1) formula*

Out[7]: `array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])`

In [8]: `np.arange(10,20)` *# It arrange the value in between 10 to 20, right side value use*

Out[8]: `array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])`

In [9]: `np.arange(10,50,5)`

Out[9]: `array([10, 15, 20, 25, 30, 35, 40, 45])`

In [10]: `np.arange(10,20,30,5)` *# It arrange only 3 argument*

```

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TypeError                                Traceback (most recent call last)
Cell In[10], line 1
----> 1 np.arange(10,20,30,5)

TypeError: Cannot interpret '5' as a data type

```

```
In [11]: np.arange(20,10)    # In arrange function 1st argument always less than 2nd argume
```

```
Out[11]: array([], dtype=int64)
```

```
In [12]: np.arange(-20,10)    # It display the value -20 to +10 value like negative to posi
```

```
Out[12]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
               -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
                6,  7,  8,  9])
```

Zeros function

```
In [13]: np.zeros(3)    # 1D array in float that is called parameter tuning
```

```
Out[13]: array([0., 0., 0.])
```

```
In [46]: np.zeros(3, dtype=int)    # 1D array in float that is called hyperparameter tunnin
```

```
Out[46]: array([0, 0, 0])
```

```
In [15]: np.zeros([2,2])    # This function display 2 by 2 matrix in '0' float value
```

```
Out[15]: array([[0., 0.],
               [0., 0.]])
```

```
In [16]: np.zeros([10,10])    # This function display 10 by 10 matrix in '0' float value
```

```
Out[16]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

Ones function

```
In [17]: np.ones(4)    # It display all value converted into the '1' float value .
```

```
Out[17]: array([1., 1., 1., 1.])
```

```
In [18]: np.ones(4,dtype=int)    # It display all value converted into the '1' integer valu
```

Out[18]: array([1, 1, 1, 1])

In [19]: `np.two(2,3)` *# Here Two is not define as function in numpy it show error.*

```
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AttributeError                                Traceback (most recent call last)
Cell In[19], line 1
----> 1 np.two(2,3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'two'
```

Random,randint

In [20]: `np.random.rand(4)` *# it arrange the value randomly in between 0 to 4*

Out[20]: array([0.66677275, 0.3843314 , 0.55862188, 0.38249353])

In [21]: `np.random.rand(3,4)` *# it arrange the value randomly in between 3 to 4 everytime i*

Out[21]: array([[0.82748002, 0.97209768, 0.33934082, 0.47736499],
 [0.90510041, 0.5008864 , 0.75137333, 0.96752866],
 [0.29650847, 0.24255968, 0.33000812, 0.17335695]])

In [28]: `np.random.randint(3)` *# It is ineger number genrate between range (0-3) randomly*

Out[28]: 1

In [31]: `np.random.randint(10,20)` *# It is ineger number genrate between range (10,20) random*

Out[31]: 17

In [33]: `np.random.randint(1,10,4)` *# it display the element randomly in between 1 to 10*

Out[33]: array([9, 5, 4, 9], dtype=int32)

In [34]: `print(np.random.randint(1,10,4))`

[5 8 3 7]

In [36]: `m= np.random.randint(10,40,(10,10))` *# every image is create 10 by 10 matrix in be*
`print(m)`

```
[[17 24 10 35 25 28 35 18 15 39]
 [16 12 12 23 36 17 32 32 23 20]
 [36 27 36 31 26 18 17 26 11 25]
 [13 27 28 20 18 27 13 31 12 39]
 [33 18 20 11 22 13 39 17 22 35]
 [17 33 32 30 32 19 15 14 23 21]
 [33 28 16 14 33 18 35 13 34 18]
 [35 36 34 18 30 13 24 33 28 13]
 [23 31 35 39 23 26 25 25 10 37]
 [39 21 28 23 38 13 27 28 37 30]]
```

```
In [37]: m= np.random.randint(1,10,(4,6))    # It select random element in between 10 to 40,w
print(m)
```

```
[[3 2 5 9 3 2]
 [2 4 4 9 4 5]
 [5 5 3 4 6 7]
 [9 9 6 8 6 5]]
```

indexing and slicing in array

```
In [38]: print(m)
```

```
[[3 2 5 9 3 2]
 [2 4 4 9 4 5]
 [5 5 3 4 6 7]
 [9 9 6 8 6 5]]
```

```
In [39]: m[0]
```

```
Out[39]: array([3, 2, 5, 9, 3, 2], dtype=int32)
```

```
In [43]: m[-3]    # it display the -3 row it start from last start down to up.
```

```
Out[43]: array([2, 4, 4, 9, 4, 5], dtype=int32)
```

```
In [44]: print(m[0,5])    # It display specific single value in row
```

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```
In [45]: print(m[-2,-4])    # It display the value -2 row in which it pick -4 value element.
```

```
3
```

Numpy operations

```
In [47]: np.arange(0,10)
```

```
Out[47]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

Reshape function

```
In [48]: arr= np.arange(1,10)           # In this Reshape function it arrange the element in
        print(arr.reshape(3,3))

[[1 2 3]
 [4 5 6]
 [7 8 9]]
```

```
In [49]: a = np.random.randint(0,100,10)    # In this operation it display the random any va
        a
```

```
Out[49]: array([36, 11, 19, 45, 84, 55, 78, 73, 12, 24], dtype=int32)
```

Seed

```
In [51]: np.random.seed(12)
```

```
In [53]: arr= np.random.randint(0,100,10)  # Here you can not get the random values, it show
        arr
```

```
Out[53]: array([75, 27,  6,  2,  3, 67, 76, 48, 22, 49], dtype=int32)
```

```
In [57]: arr2 = np.random.randint(0,100,10)    # whenever we exucute output it shows differa
        arr2
```

```
Out[57]: array([59, 89, 59, 88, 79, 25, 97, 56,  4, 96], dtype=int32)
```

```
In [58]: arr2
```

```
Out[58]: array([59, 89, 59, 88, 79, 25, 97, 56,  4, 96], dtype=int32)
```

Max

```
In [59]: arr2.max()           # It pick one maximum value form arr2 elements.
```

```
Out[59]: np.int32(97)
```

mini

```
In [62]: arr2.min()           # It pick one minimumm value form arr2 elements.
```

```
Out[62]: np.int32(4)
```

Mean

```
In [63]: arr2.mean()           # It calulate mean of given arr2 elements
```

```
Out[63]: np.float64(65.2)
```

```
In [64]: arr2.median()         # It does not calculate median,mode only numpy allow 'mean' functi
```

```
-----  
AttributeError                                Traceback (most recent call last)  
Cell In[64], line 1  
----> 1 arr2.median()  
  
AttributeError: 'numpy.ndarray' object has no attribute 'median'
```

Reshape

```
In [65]: arr2.reshape(2,5)
```

```
Out[65]: array([[59, 89, 59, 88, 79],  
               [25, 97, 56,  4, 96]], dtype=int32)
```

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